Getting to know Cisco Firewalls

Cisco has replaced its traditional Adaptive Security Appliance (ASA) firewalls with the new Firepower next-generation firewalls. For our study in this module, we have procured the Firepower 1000 series firewalls meant for small businesses and branch offices, specifically Firepower 1010 as shown in Figure 1.

Figure 1

(To add exposure for the learning benefits of students, we have also procured Palo Alto PA-410 next-generation firewalls meant for enterprise branch as shown in Figure 2 but is awaiting for delivery and mounting onto the network racks to be included into our study in future.)

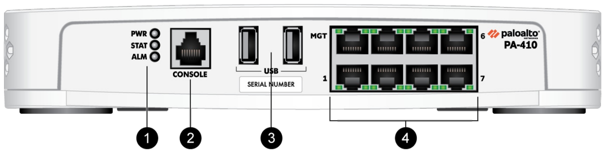

Figure 2

To be competitive in the market, next-generation firewalls (NGFW) are more than a stateful firewall. For example, Cisco firepower running FXOS (Firepower eXtensible Operating System) with Threat Defense application combines capabilities with intrusion prevention system (IPS), advanced malware protection (AMP) and threat intelligence with 24/7 online updates provided by Cisco Talos team. Obviously, all these capabilities come with additional licensing requirements and cost which are meant more for real-world industry needs rather than academic study where the firewalls are usually turned off when not in use.

Hence, to avoid unnecessary complexity and confusion in the study of firewalls especially for beginners, the Firepower firewalls in the lab have been re-imaged to run FXOS with ASA application to behave as a traditional stateful firewall.

In practice, a firewall is typically deployed in routed mode, i.e. operating similarly as a router to route network traffic. In the Firepower 1010 hardware, notice that there are eight GigabitEthernet ports labeled Ethernet 1/1 - 8 which are different from typical port labeling found in switches and routers. By default, only Ethernet 1/1 is a routed port meant to be connected to the edge router towards the Internet. Ethernet 1/2 - 8 are switch ports meant to be connected to the internal enterprise networks as shown in Figure 3.

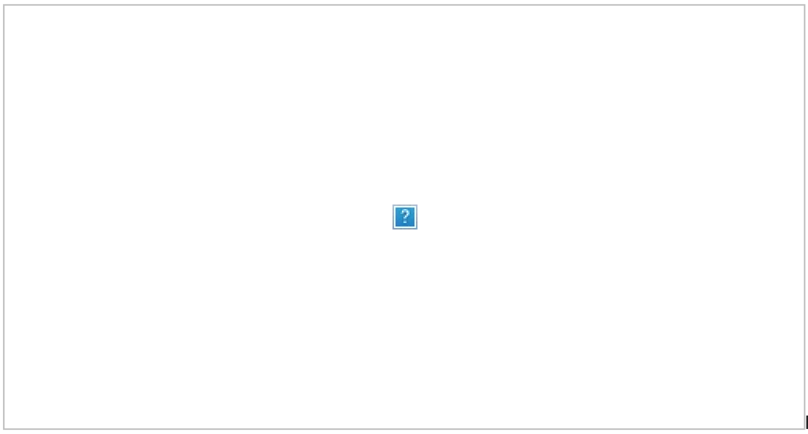

Figure 3

For a Firepower/ASA firewall to be operational, first configure the required Firepower switch port into routed port using the same command as in layer-3 switch:

```
ciscoasa(config)# interface interface-id
ciscoasa(config-if)# no switchport
```

Then, configure the following to be elaborated in details next:

1. Security level (mandatory)
2. ACLs (optional)
3. Modular Policy Framework (MPF) (additonal)

<u>1. Mandatory Configuration of Security-Level for Segmenting Networks into Security Zones</u>

At the minimal, each connected interface of a Firepower/ASA firewall must be configured with an IP address and a subnet mask. In addition, a name and a security-level are required to be configured before it can be operational. An example is shown in Figure 4.

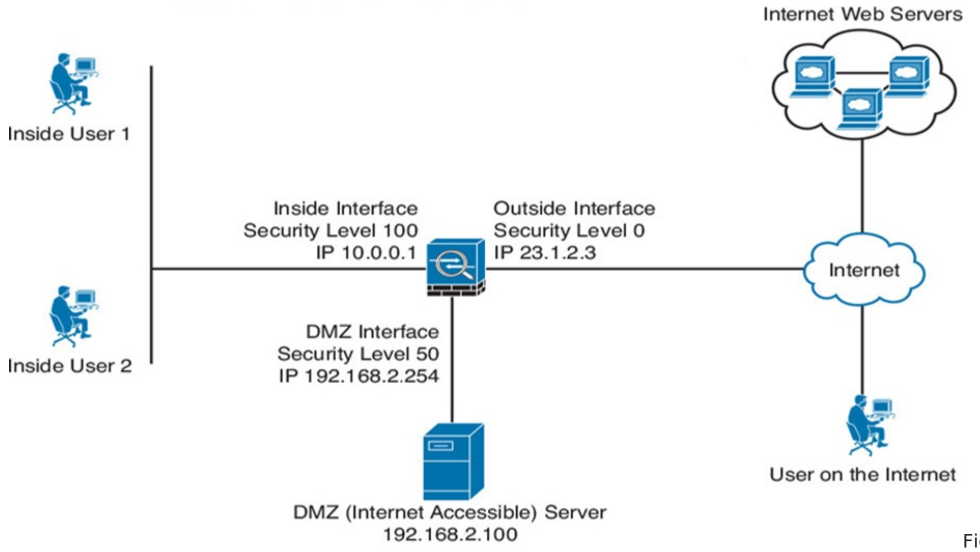

Figure 4

Commands to configure an interface of a Firepower/ASA firewall:

```
ciscoasa(config)# interface interface-id
ciscoasa(config-if)# ip address ip-addr subnet-mask
ciscoasa(config-if)# nameif interface-name
ciscoasa(config-if)# security-level num-between-0-100
```

Security-level configured with a number between 0 to 100 inclusive is to define a security zone. 0 represents least trusted security zone typically meant for the interface connected to the Internet whereas 100 represents most trusted security zone typically meant for interface connected to the internal enterprise LAN. Interface connected to DMZ may be configured with a number in between, say 50. In fact, these numbers are relative and any suitable number may be configured.

By default, network traffic initiated from higher security-level to lower security-level are allowed to pass through the firewall. In contrast, network traffic initiated from lower security-level to higher security-level will be denied and dropped as shown in Figure 5.

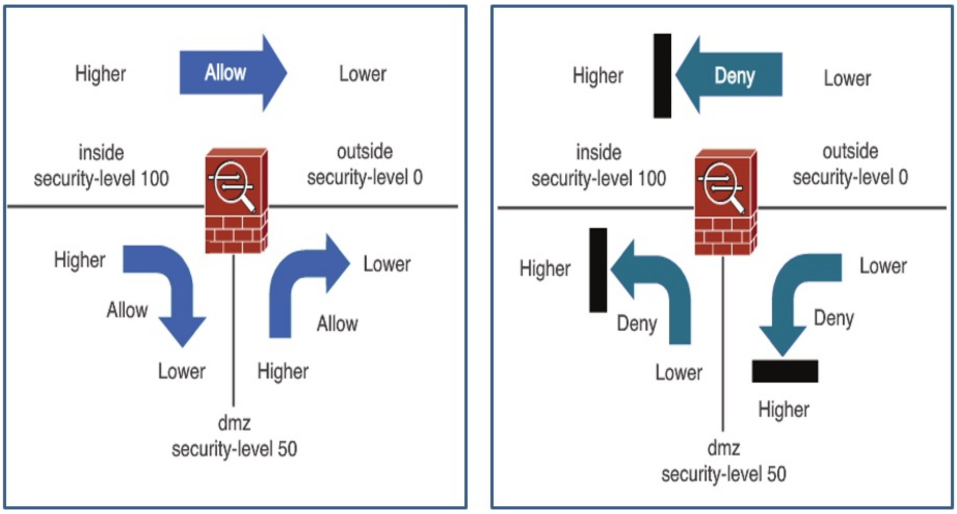

Figure 5

However, to support 2-way communications, note that return network traffic from lower security-zone to higher security-zone are allowed by default as shown in Figure 6. This is achieved by maintaining a state table or connection table within the firewall. Hence, a firewall is also known as a stateful inspection firewall. In fact, a router configured with ACLs can be theoretically considered as a stateless or packet-filtering firewall due to no state table.

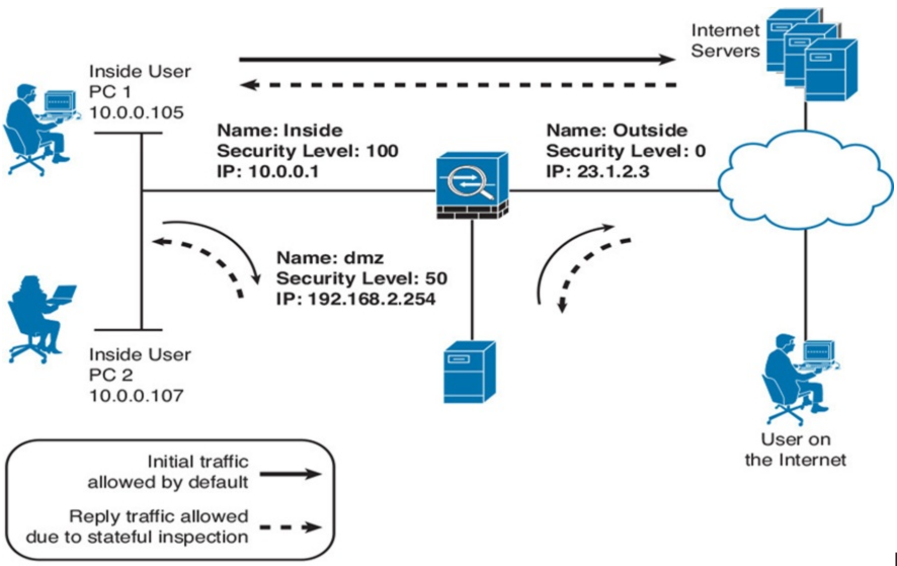

Figure 6

Figure 7 shows an example state table maintained in the Firepower/ASA firewall.

```
ciscoasa# show conn
352 in use, 5985 most used
UDP DMZ 172.16.0.25:161 inside 10.0.0.30:1879, idle 0:00:27, bytes 706509, flags -
UDP outside 192.0.2.18:123 inside 10.0.0.108:123, idle 0:00:34, bytes 79, flags -
TCP outside 192.0.2.150:80 DMZ 172.16.0.5:59512, idle 0:00:00, bytes 0, flags Uf
TCP outside 192.0.2.150:80 inside 10.0.0.102:59559, idle 0:00:03, bytes 1488,
   flags UfFRIO
...
```

Figure 7

If for any reason you are required to configure two interfaces of a Firepower/ASA firewall with the same security-level, e.g. dmz1 and dmz2 as shown in Figure 8, note that by default network traffic are not allowed to pass between them.

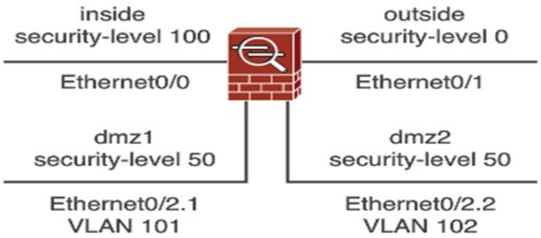

Figure 8

To override default security policy to allow network traffic between 2 interfaces with the same security-level, you will need to configure:

```
ciscoasa(config)# same-security-traffic permit inter-interface
```

Note that in routed-mode, a firewall will need to operate together with routers and layer-3 switches in an enterprise LAN. An example is shown in Figure 9.

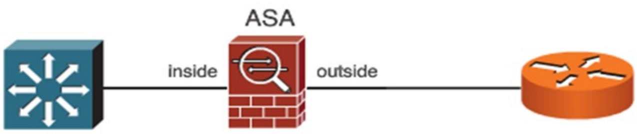

Figure 9

Hence, remember to also configure static and default routes, or OSPF where necessary. The command for configuring static route:

```
ciscoasa(config)# route int-name subnet-block netmask next-hop-ip
```

For help in configuring Firepower/ASA firewall with OSPF, you may read: [How to Configure OSPF on Cisco ASA Firewall]

NAT is typically configured on the edge router. If you wish to configure NAT on Firepower/ASA firewall instead of edge router, you may read: [Cisco Firepower & Cisco ASA - NAT Configuration Guide]

2. Optional Configuration of ACLs for Overriding Default Security Policy

Recall in Figure 4 that DMZ is meant for an enterprise to host servers accessible from the Internet and is configured with a security-level higher than the Internet since it is conceptually more trusted than the Internet. In this case, how can users from the Internet access the servers since network traffic initiating from the Internet with lower security-level will be denied and dropped by default?

The answer is to configure ACL to specifically permit access which in fact fulfils the recommended deny by default security principle that will minimize accidental oversight resulting in security compromise.

Consider the scenario in Figure 10 where a server with IP address 209.165.200.226 is hosted at DMZ for access from the Internet.

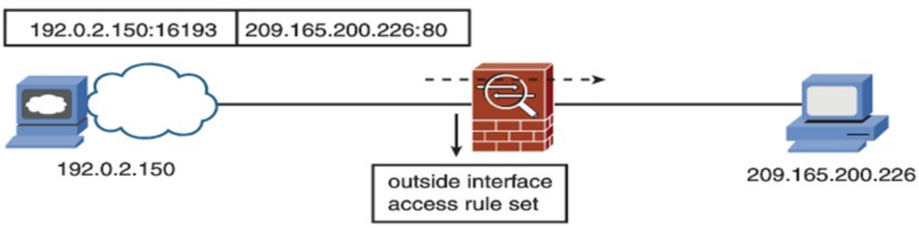

Figure 10

To permit any user, say at IP address 192.0.2.150 from the Internet to access the server, ACL will be written and applied at the outside interface connecting to the Internet. The ACL commands in Firepower/ASA firewall are similar as the routers but with some differences. An example as shown:

```
ciscoasa(config)# access-list acl-name permit tcp any host 209.165.200.226 eq www
ciscoasa(config)# access-group acl-name in interface int-name
```

Not shown in the above example, note that ACL in Firepower/ASA firewall uses subnet mask instead of wildcard mask to specify subnet block.

3. Additonal Configuration of Modular Policy Framework (MPF) for Other Security Features

Essentially, configuring MPF involves 3 steps as shown in Figure 11.

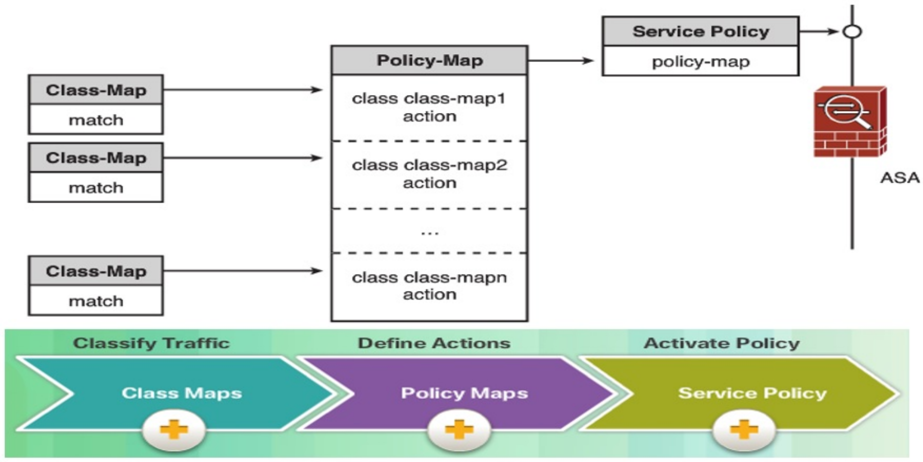

Figure 11

1. Configure class-map to specify network packets to be matched. This is similar in concept to configuring test condition in ACL.
2. Configure policy-map to specify the action to take on the packets matching the class-map. This is similar in concept to configuring the permit/deny action in ACL, but policy-map allows other actions to be discussed shortly.
3. Configure service-policy to apply the policy-map. This is similar in concept to configuring access-group in ACL.

For illustration, Figure 12 shows the default configuration of class-map, policy-map and service-policy in the running-config of a Firepower/ASA firewall.

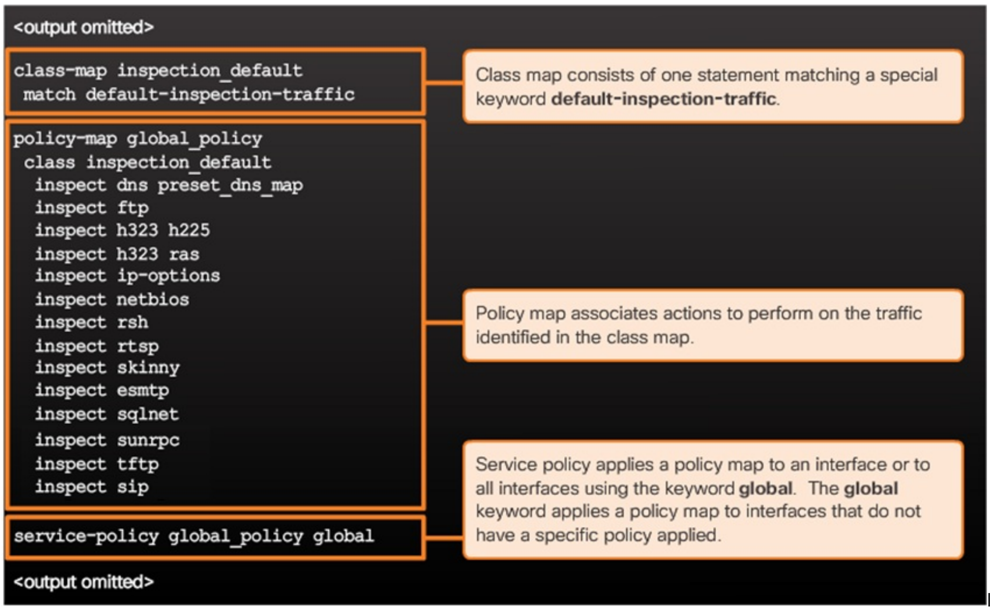

Figure 12

In the class-map configuration, default-inspection-traffic is a special keyword for matching network packets typically expected to be inspected in practice as shown in Figure 13.

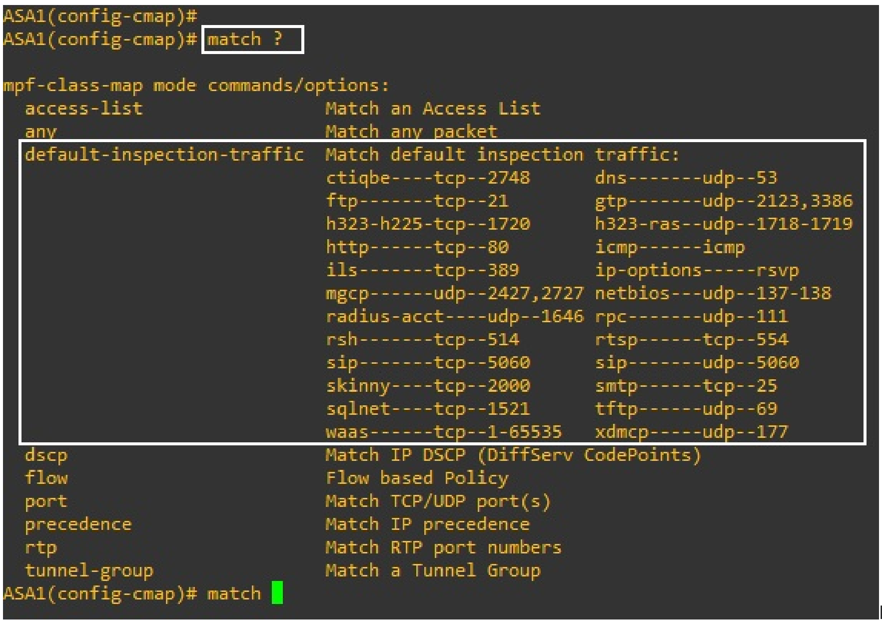

Figure 13

In the policy-map configuration, inspect is the keyword to enable the respective inspection engines in the Firepower/ASA firewall to inspect the corresponding network packets.

In the security-policy configuration, global is the keyword to apply the policy-map globally on all interfaces of the Firepower/ASA firewall. Note that only one policy-map can be applied globally. Each interface of a Firepower/ASA firewall may also be applied with one specific policy-map which will take precedence over the global policy.

To have better understanding, let's explore more details of configuring MPF in a Firepower/ASA firewall:

- Modifying default global MPF

- Configuring specific interface MPF

3A. Modifying Default Global MPF

For security reason, a Firepower/ASA firewall will drop ping packets by default. Nevertheless, it can be configured to permit ping packets by modifying the default global MPF as follows:

```
ciscoasa(config)# policy-map global_policy
ciscoasa(config-pmap)# class inspection_default
ciscoasa(config-pmap-c)# inspect icmp
```

Note that `global_policy` and `inspection_default` are the names referring to the default class-map and policy-map in Figure 12. The `inspect icmp` command is to enable the icmp inspection engine. Without it in the default configuration, ping request initiated from higher security zone will not be monitored in the state table. As a result, ping reply from lower security zone will be denied and ping will fail.

3B. Configuring Specific Interface MPF

Figure 14 shows the command syntax for configuring a class-map:

```
ciscoasa(config)# class-map class_map_name
ciscoasa(config-cmap)# description text
ciscoasa(config-cmap)# match ...
ciscoasa(config-cmap)# exit
```

Figure 14

Figure 15 shows some examples of configuring the match command which you can practice in subsequent lab exercises:

Matching Condition	Command Syntax
Any traffic	ciscoasa(config-cmap)# **match any**
Default traffic types	ciscoasa(config-cmap)# **match default-inspection-traffic**
Destination port number	ciscoasa(config-cmap)# **match port** {**tcp** | **udp**} {**eq** *port* | **range** *start end*}
Access list	ciscoasa(config-cmap)# **match access-list** *acl_name*

Figure 15

Note that `match default-inspection-traffic` is configured in the default class-map.

Figure 16 shows the command syntax for configuring policy-map:

```
ciscoasa(config)# policy-map policy_map_name
ciscoasa(config-pmap)# description text
ciscoasa(config-pmap)# class class_map_name
ciscoasa(config-pmap-c)# ...
```

Figure 16

Note that the command `class class_map_name` is to insert the class-map configured in Figure 14 so that configured actions will be taken on the traffic matching the class-map.

Figure 17 shows some examples of configuring the actions to take on the matched traffic in the policy-map which you can practice in subsequent lab exercises:

Action	Command Syntax
Set connection limits	ciscoasa(config-pmap-c)# **set connection ...**
Inspect applications	ciscoasa(config-pmap-c)# **inspect** *engine_name*
Police the traffic	ciscoasa(config-pmap-c)# **police** {**output** | **input**} *conform_rate* [*burst_bytes*] [**conform-action** {**drop** | **transmit**}] [**exceed-action** {**drop** | **transmit**}]
Apply priority handling	ciscoasa(config-pmap-c)# **priority**

Figure 17

Note the various `inspect engine_name` configured in the default policy-map.

Finally, Figure 18 shows the command syntax of service-policy to apply the policy-map either globally or locally on specific interface:

```
ciscoasa(config)# service-policy policy_map_name {global | interface if_name}
```

Figure 18

Packet Processing Order of a Firepower/ASA Firewall

Last but not least, it's instructional to understand how a Firepower/ASA firewall process a packet. Figure 19 shows the order of processing when a packet arrives at the ingress interface of a Firepower/ASA firewall in Step 1.

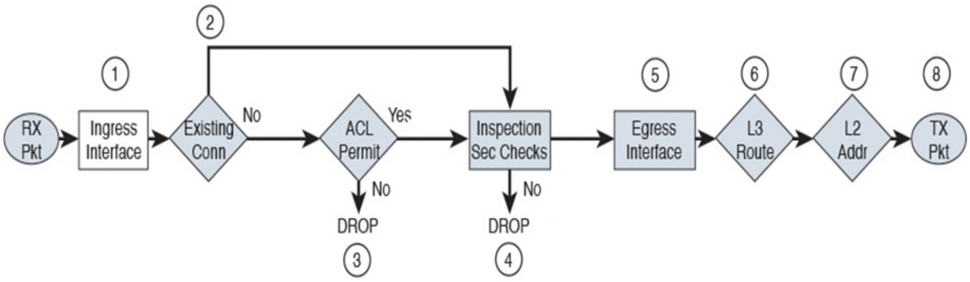

Figure 19

First, the packet will be checked in Step 2 against the state table maintained by the firewall after the configuration of security-level discussed earlier. If it is a return packet matching an existing connection, it will bypass Step 3 ACL check and go directly to Step 4 for required inspection security check if configured using MPF.

If the packet passes the security policy, it will be routed to the egress interface in Step 5 for onward transmission to the destination.